

F. Comfortably Numb

time limit per test: 8 seconds
memory limit per test: 1024 megabytes

You are given an array a consisting of n non-negative integers.

The *numbness* of a subarray $a_l, a_{l+1}, \dots, a_r$ (for arbitrary $l \leq r$) is defined as

$$\max(a_l, a_{l+1}, \dots, a_r) \oplus (a_l \oplus a_{l+1} \oplus \dots \oplus a_r),$$

where $\oplus$ denotes the [bitwise XOR operation](#).

Find the maximum numbness over all subarrays.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 1000$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print one integer — the maximum numbness over all subarrays of the given array.

Example

input

Copy

2
5
1 2 3 4 5
3
10 47 52

output

Copy

7
47

Note

For the first test case, for the subarray $[3, 4, 5]$, its maximum value is 5. Hence, its numbness is $3 \oplus 4 \oplus 5 \oplus 5 = 7$. This is the maximum possible numbness in this array.

In the second test case the subarray $[47, 52]$ provides the maximum numbness.

Codeforces Round 845 (Div. 2) and ByteRace 2023

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
390193040	Sep/10/2026 13:46	Accepted
390191660	Sep/10/2026 13:30	Wrong answer on test 2

→ Problem tags

bitmasks

data structures

divide and conquer

strings

trees

*2400

No tag edit access

→ Contest materials

Announcement (en)

Tutorial (en)